Guided Generation of Cause and Effect

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2021

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Why Open Causal Generation?

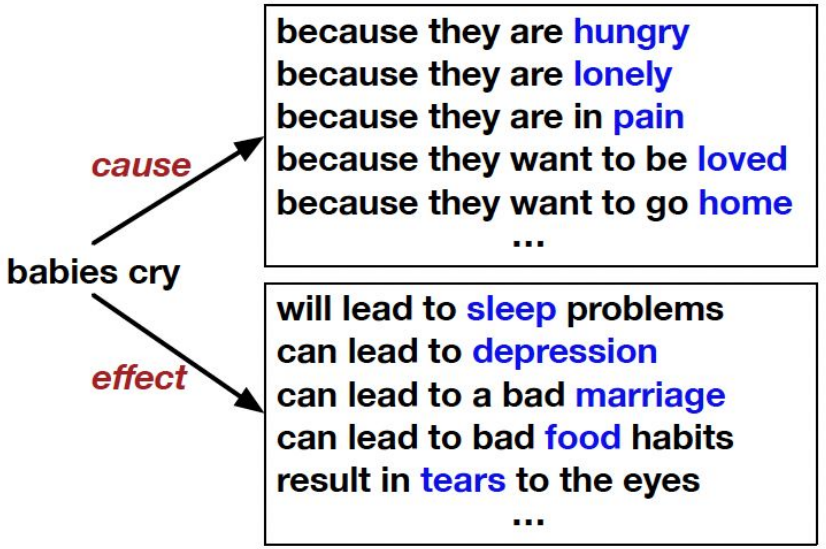
Goal: Given any free-form event, generate multiple plausible causes and effects.

Open generation

- The model generates its own natural-language possibilities instead of choosing from a fixed answer list.

Main challenges

1. Need massive causal training data
2. Need outputs that are diverse in meaning not just paraphrase
3. Need to guide generation while keeping sentences natural



Example from the paper: “babies cry” can have multiple plausible causes and many plausible effects.

Key Idea and Contributions

CausalBank

314M sentence-level cause-effect pairs mined from Common Crawl

Cause Effect Graph

89.1M lexical causal edges used to find useful constraint keywords.

DPC Decoding

Disjunctive Positive Constraints let the decoder use one acceptable word form.

Evaluation

Human Evaluation + COPA show quality, diversity and useful causal knowledge

➔ **Causal Bank teaches -> CEG guides -> DPC constrains flexibility**

Overall Approach

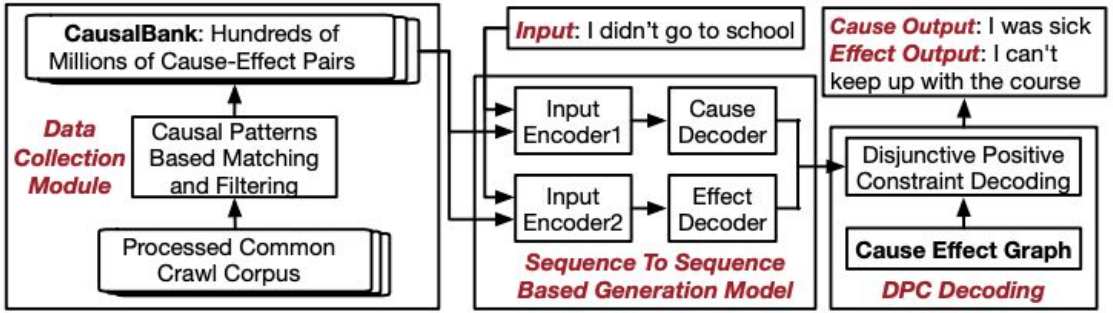


Figure 2: Our approach for generating plausible causes and effects.

Causal Pattern
as, as a consequence/result of, as long as, because, because of, caused by, due/owing to, in response to, on account of, result from
accordingly, consequently, bring on/about, give rise to, induce, in order to, lead to, result in, prevent/stop...from, and for this reason, cause, for the purpose of, if...then, ,_so, so that, thereby, therefore, thus, hence

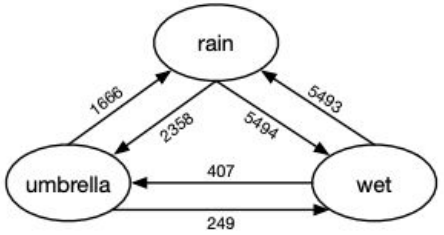


Figure 3: Cause Effect Graph: A lexical causal knowledge base.

Fig 2: The system combines large-scale causal data, two sequence-to-sequence generation models, a lexical Cause Effect graph, and DPC Decoding.

2.1 CausalBank: Building the Training Data

1. Start with Web text

5.14 TB processed Common Crawl corpus.

2. Match causal patterns

Because, due to, result in, lead to, therefore, if..then, etc

3.Identify direction

EPC: Effect-Pattern-Cause
CPE: Cause-Pattern-Effect

4. Filter

Remove duplicates, negated relations, passive voice, require >=2 tokens

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EPC example

“I am very sad BECAUSE I lost my phone.”
Effect -> Pattern -> Cause

CPE example

“The earthquake RESULTED IN many deaths.”
Cause -> Pattern -> Effect

133M EPC + 181M CPE = 314 MILLION cause-effect pairs

Manual check of 1000 samples: 95% conveyed a meaningful causal relation.

2.2 Cause Effect Graph (CEG)

What is it?

A lexical (word-level) causal knowledge base. Nodes are lemmatized terms; directed edges represent causal relations.

How is it built?

From causal sentences in Common Crawl. Stop words are removed; nouns; verbs, adjectives and adverbs are retained. Edge weights reflect co-occurence frequency.

Example

“The storm caused… damage…” -> lexical pairs such as storm -> damage, storm -> beach.

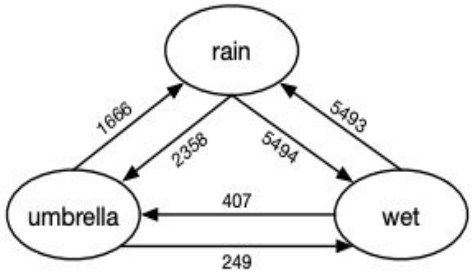


Figure 3: Cause Effect Graph: A lexical causal knowledge base.

CEG size: 89.1M causal edges

CausalBank = trains the generator
CEG = guides the generator

2.3 Guided Generation: Why DPC?

Problem: a CEG constraint is stored as a lemma, but natural sentences need the right morphological form.

Input

"I bought an umbrella."

CEG suggests the causal concept: RAIN

Old positive constraint

Force the exact form "rain."

Too restrictive: natural output may need "rained" or "rainy"

DPC

Require ONE of:{ rain, rains, rained, raining} related morphological forms.

Possible output: "It rained."

Disjunctive = OR ; Positive constraint = at least one allowed sequence must appear.

How DPC Supports Diverse Generation

Different CEG keywords push the same input toward different causal ideas.

Input

Babies cry

CEG constraint: hungry

Because they are hungry

CEG constraint: lonely

Because they are lonely

CEG constraint: pain

Because they are in pain

CEG constraint: love

Because they want to be loved

CEG chooses WHAT concept to explore. DPC controls HOW that concept can appear naturally.

Generation and Output Reranking

For each Input, the system creates many guided candidates and then ranks them.

Lemmatize Input	e.g., baby, cry
Query CEG	take high-frequency causal candidate lemmas
Create DPC set	generate morphological variants for each candidate
Decode	N = 300 constraints; keep M =5 outputs each
Rerank	rank NxM candidates by negative log-likelihood and return top K

Purpose: prevent a few preferred constraints from dominating the N-best outputs and encourage broader causal possibilities

Evaluation: Does it Work?

Human evaluation

Methods compared: KNN, GPT-2, random sampling, N-best, CEG constraints (CN-Cons), and gold constraints.

Main generation result

On unseen COPA examples, constrained methods remain strong while KNN drops sharply, CN-Cons also generally gives the greatest lexical diversity.

CausalBERT / COPA

Adding CausalBank intermediate training improves BERT-base COPA test accuracy from 75.4% -> 78.6%(+3.2 points)

	Method	Cause				Effect			
		P@1	P@3	H	Div	P@1	P@3	H	Div
TrainSub	KNN	**89.0**	67.3	**0.85**	0.11	**98.0**	71.3	**0.90**	**0.02**
	GPT-2	31.0	22.3	0.39	0.13	8.0	9.3	0.30	0.11
	N-Best	59.0	45.3	0.53	0.15	63.0	42.7	0.53	0.11
	Random	68.0	59.3	0.66	0.11	74.0	61.7	0.70	0.09
	CN-Cons	72.0	71.3	0.79	**0.02**	66.0	67.0	0.76	**0.02**
	Gold-Cons	78.0	**75.3**	0.83	0.12	71.0	**73.0**	0.80	0.10
COPA_Dev	KNN	10.0	8.0	0.53	0.10	4.0	2.7	0.26	**0.01**
	GPT-2	40.0	34.0	0.45	0.12	38.0	32.0	0.46	0.10
	Random	66.0	53.7	0.65	0.09	62.0	46.7	0.57	0.08
	N-Best	69.0	65.0	0.77	0.08	**72.0**	68.0	0.82	0.07
	CN-Cons	**74.0**	70.0	0.81	**0.02**	**72.0**	**72.0**	**0.87**	0.02
	Gold-Cons	73.0	**73.0**	**0.87**	0.09	**72.0**	71.3	**0.87**	0.09

Table 3: Human evaluation results of cause and effect generation.

The goal of the causalBERT experiment is to show that CausalBank contains useful causal knowledge - not to beat the largest T5 model.

Strengths and Limitations

Strengths

- Introduces open-ended cause/effect generation
- Very large automatically mined resource: 314M pairs
- CEG +constrained decoding improves semantic diversity
- DPC is reusable beyond causality (e.g., synonyms or alternate segmentations)
- CausalBank improves downstream causal reasoning

Limitations

- Focuses on explicitly expressed causal relations
- Pattern-based extraction can contain noise
- Web-mined data can suffer from reporting bias
- Generated causes/effects are plausible possibilities, not verified real-world causes
- Human evaluation agreement is moderate (K = 0.53)

Future Work and Extensions

Authors' Future Directions

1. Better Causal Extraction

Improving corpus preprocessing and cause/effects span extraction instead of relying only on fixed pattern boundaries.

2. Better continued-training strategies

Explore adaptation methods that preserve pretrained knowledge when adding CausualBank data.

3. Broader use of DPC

Apply disjunctive constraints to synonyms or alternative subword segmentations, not only morphological variants.

My Proposed Extensions:

1. Implicit causal relations

Model causal relationships even when explicit markers such as "because," "due to," or "result in" are absent.

2. Context-aware generation

Use multi-sentence or event context so generated causes and effects fit the surrounding situation.

3. Modern LLM + structured causal knowledge

Combine CEG-style retrieval or causal knowledge with modern language models for more grounded generation.

Key Takeaways

CausalBank - TEACHES

314M cause-effect pairs

Provides large-scale sentence-level training data.

Cause Effect Graph - GUIDES

89.1M lexical causal edges

Supplies useful causal concepts that can guide different generations.

DPC - CONSTRAINTS FLEXIBLY

One of several forms can satisfy a constraint.

Lets the model use a natural morphological form while preserving lexical guidance.

CausalBank teaches -> CEG guides -> DPC constrains flexibility -> diverse causal generation

References

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